

TANISHK GANGWAR

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EDUCATION

Manipal University Jaipur

Oct 2024 – Jun 2028 (Expected)

Bachelor of Technology, Computer Science & Engineering (Data Science)

- CGPA: 9.00/10.
- Dean's Academic Excellence Award, three consecutive semesters (Sem 2, 3 & 4).
- Relevant Coursework: Data Structures & Algorithms, Operating Systems, Computer Networks, DBMS, OOP, Cloud Computing, Machine Learning, Data Mining.

EXPERIENCE

Vocational Trainee – Advanced Pattern Recognition

Jun 2026 – Jul 2026

NTPC Limited

- Eliminated recurring manual file-reformatting work on a locked-down office machine with no Python or library access by building a standalone, fully offline browser-based converter (HTML + SheetJS) that reformats industrial Trend Report exports in-browser.
- Replaced manual, plot-by-plot outlier review in the team's Pearson correlation analysis by building an automated pipeline that removes outliers via IQR/MAD-based modified z-scores and flags features for rejection based on non-continuous plot patterns.
- Shipped the CSV converter's production version after 7 iterative rounds with the APR team, refining format handling and edge cases based on direct feedback against their actual Trend Report workflow.

Open Source Contributor

pandas (*pandas-dev/pandas*), *PyDriller* (*ishepard/pydriller*), *PyGithub* (*PyGithub/PyGithub*)

- Fixed a data-integrity bug in pandas (200M+ monthly downloads) by rewriting negative-slice indexer validation with `slice.indices()` (PR #66101), shipping regression tests guarding core indexing behavior.
- Restored documented diff-counting behavior in PyDriller by fixing `Commit._stats()` to respect the `skip_whitespace` flag (PR #320), verified with an expanded unit test suite covering both flag states.
- Closed a long-standing PyDriller feature request by implementing a `Commit.patch` property exposing full unified diffs (PR #321), enabling patch-level analysis for repository-mining pipelines, validated with new test cases.
- Cut worst-case client stalls from unbounded waits to a configurable ceiling in PyGithub by adding a `max_rate_limit_wait` cap and a new `RateLimitExceededExceedsMaxWait` exception to `GithubRetry` (PR #3540).

PROJECTS

Kansei - Concurrent ML Inference System, Deployed on GCP | Python, FastAPI, Docker, CLIP, GCP, SQL ([GitHub](#), [Live Site](#))

- Engineered a concurrent FastAPI backend scoring users against 16 aesthetic centroids in 512-dimensional CLIP embedding space, cutting quiz load to a single network round-trip via async model preheating and parallelized API calls.
- Reduced Docker image size 82% (9.2GB to 1.62GB) by isolating a CPU-only PyTorch/CLIP build and splitting classification into a separate authenticated HTTP service; eliminated OOM kills of the 918MB-peak model on a 1GB-RAM GCP e2-micro via a provisioned swap file.
- Restored full production service post-migration by debugging three failures: insecure-context `crypto.randomUUID()` errors, a missing service-auth env var, and a UMAP minimum-sample crash, through log-driven root-cause isolation.

Ikiru - Staggered-Adoption Causal Study of CODEOWNERS on GitHub | Python, BigQuery, SQL, Pandas, Statsmodels ([GitHub](#))

- Built a causal-inference pipeline over 21TB of GitHub Archive data with BigQuery SQL, constructing an 89-repository panel and applying a Sun-Abraham staggered difference-in-differences design to measure CODEOWNERS' effect on PR review latency.
- Prevented a false-positive headline result by tracing two late-window coefficients to cohort confounding via split-sample analysis of a bimodal coverage distribution, reporting a defensible null effect instead of the raw estimate.

Sekivara - Causal Analysis of GitHub Copilot's Behavioral Fingerprint | Python, Pandas, Statsmodels ([GitHub](#))

- Measured GitHub Copilot's causal effect on commit structure across 638,000+ commits across 10 repositories via difference-in-differences, validating identification with a joint pre-trend F-test across all headline features (min $p = 0.29$) and HC3 robust errors.
- Scaled commit extraction to 235K-commit repositories (llvm-project) by replacing paginated GitHub API calls, which caused connection resets, with direct git-log parsing, consolidating pipeline constants into a single config source.
- Recovered a masked treatment effect ($\beta = -0.216$) by filtering automated-commit outliers contaminating the panel, separating genuine developer behavior shifts from bot and tooling artifacts.

Bias-Aware ML Pipeline - Fair Data Preprocessing System | Python, Scikit-learn ([GitHub](#))

- Built a measurement-first bias-mitigation pipeline for Random Forest models, reducing Demographic Parity Gaps by 64.5% and Equal Opportunity TPR Gaps by 47.8% while maintaining model accuracy, in collaboration with Dr. Chirag Joshi.

TECHNICAL SKILLS

Languages: Python, C++, Java, SQL, JavaScript

Frameworks: FastAPI, Flask, PyTorch, Scikit-learn, Pandas, NumPy

Developer Tools: Git, GitHub, pytest, Docker, Google Cloud Platform (Compute Engine, BigQuery), REST APIs, Linux/Bash, SQLite

Libraries: Statsmodels, linearmodels, CLIP, UMAP

Practices: Agile/Iterative Development, SDLC

ACHIEVEMENTS

- Ranked Top 1,500 of 100,000+ participants, Google "The Big Code" competitive programming challenge.
- Selected for Amazon ML Summer School, Top 3,000 of 134,421+ applicants (~2.2% selection rate).